

Unit 2 Supervised Learning- Regression & Classification:

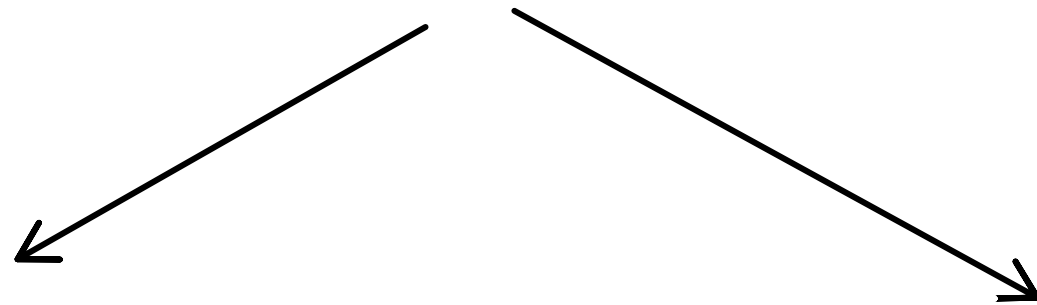
Agenda

Regression techniques including Simple Linear Regression,
understanding linear equation,
slope and intercept,

Multiple Linear Regression,
and Polynomial Regression

Evaluation of models using error metrics such as MAE,
MSE,
RMSE,
and R^2 Score.

Machine Learning



Supervised

Unsupervised

Regression

Regression: Value

Linear Regression

- Statistical Technique to find relationship between the variables
- Features and Labels

Eg:

1. Student Performance Data

Medu, FJob, NoHrsStudy, NoHrsSleep,	Marks_in_subject
Features	Label

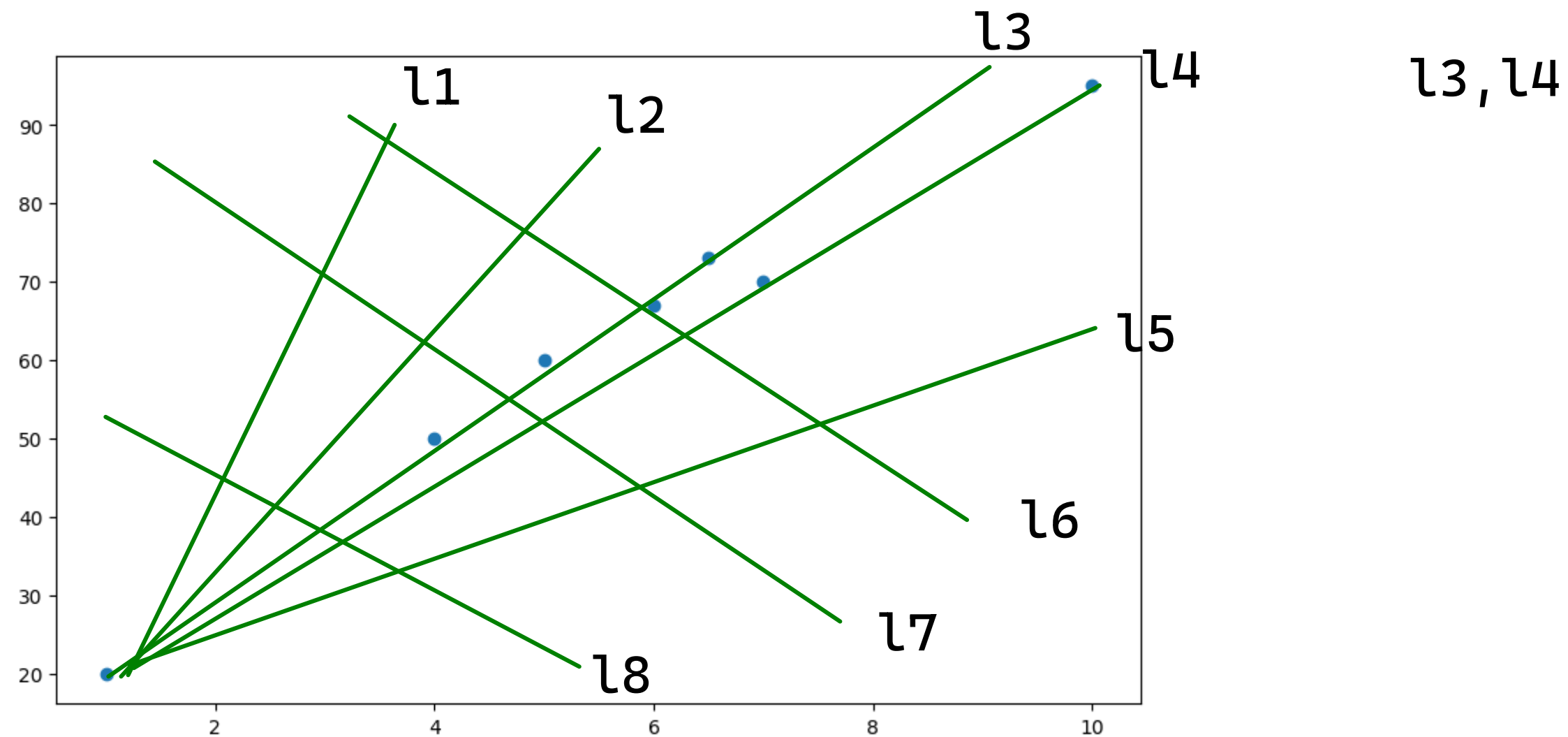
2. House Price Prediction

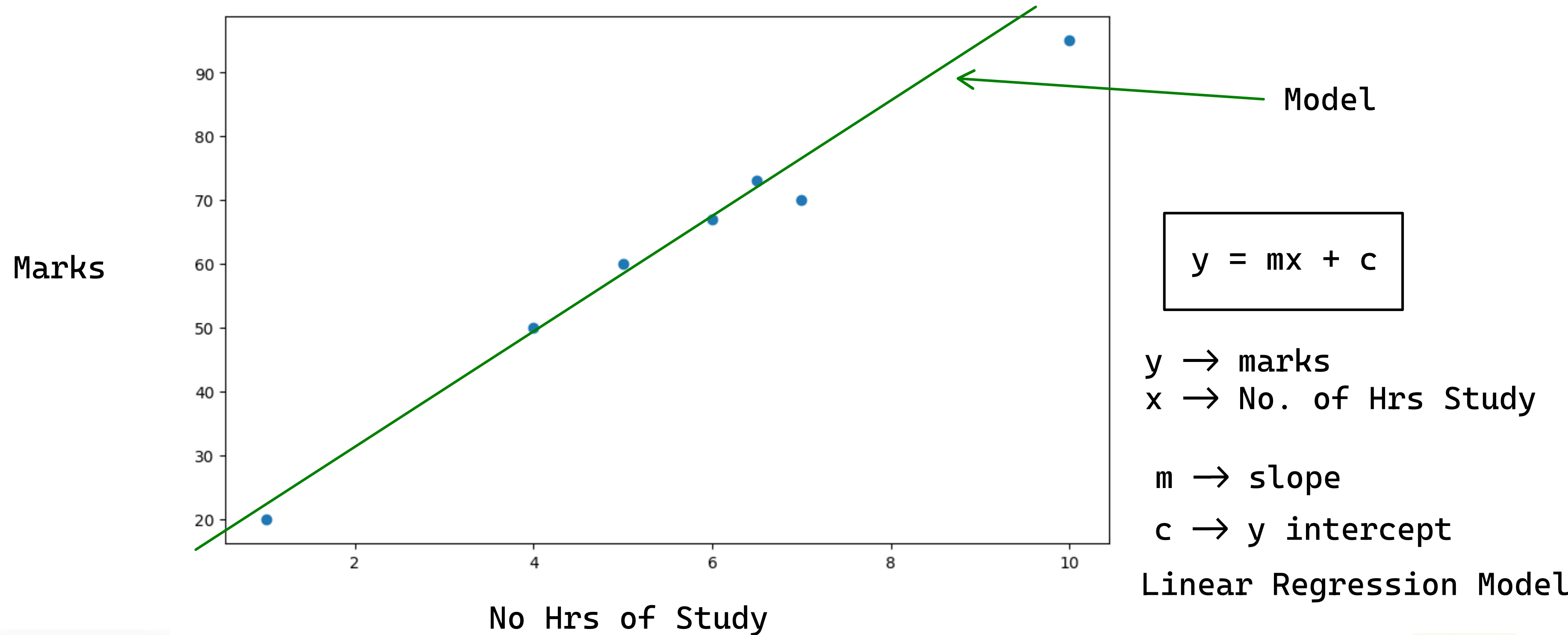
Locality, NoBedRooms, NoBathrooms, Carpet_area	Price of house
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3. Fuel Efficiency

No. of Liters of fuel, Driven Kms	Km/ltr
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Draw a line such that it touches most of the points on the data





$$m = \frac{N * \text{sum}(XY) - \text{sum}(X) * \text{sum}(Y)}{N * \text{sum}(x^2) - (\text{sum}(x))^2}$$

$$c = \frac{\text{sum}(Y) - m * \text{sum}(X)}{N}$$

Simple Linear Regression Model

$$y = mx + c$$

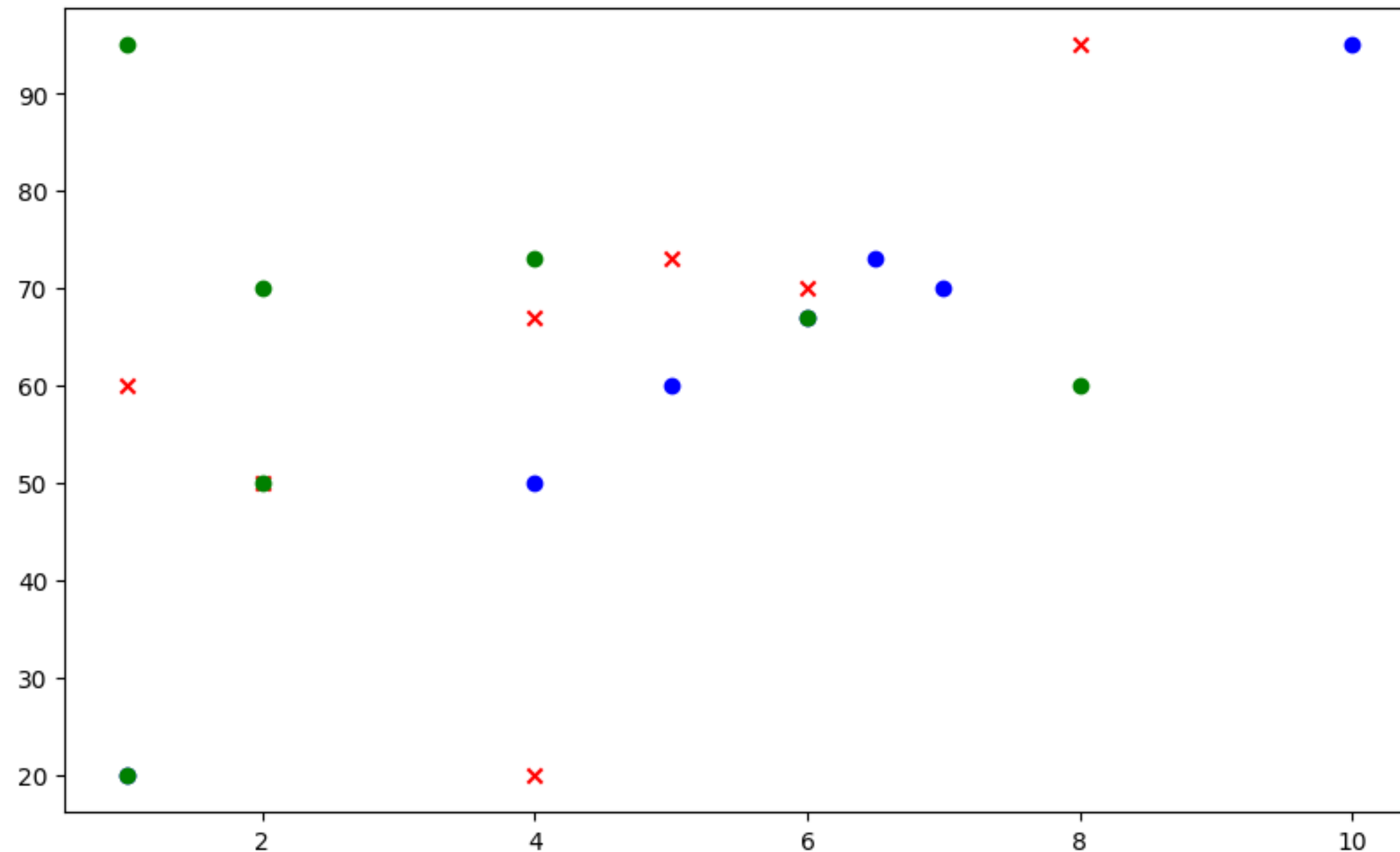
$m \rightarrow$ slope

$c \rightarrow$ intercept

Multiple Linear Regression Model: multiple independent input features

$$y = c + m_1x_1 + m_2x_2 + m_3x_3 + \dots + m_nx_n$$

$m_i \rightarrow$ slope of a single feature, keeping all other features constant



In Real time the relationship between the features and label is usually non linear

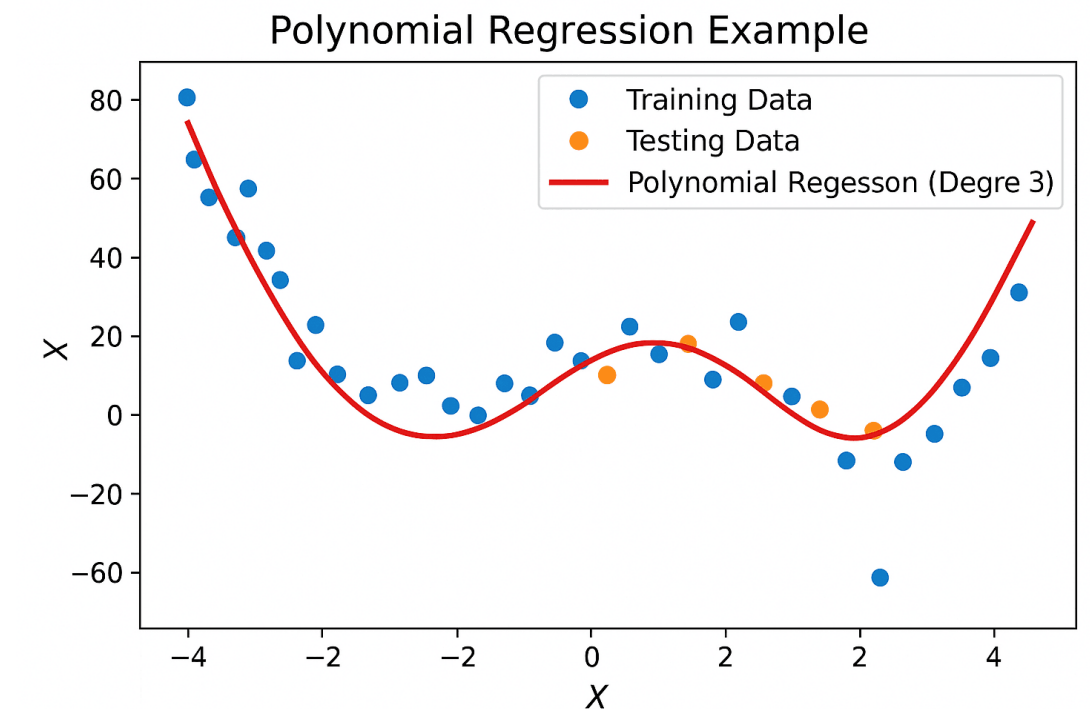
Considering a straight line results in underfitting (bias)

Polynomial regression

which adds curvelinear component to the equation

$$y = c + m_1x + m_2x^2 + m_3x^3 + \dots + m_nx^n$$

It is still considered linear regression because it is linear with respect to its parameters



What parts of the linear regression equation are updated during training?

- A. The Prediction
- B. The Bias and Weights
- C. The Feature Values

y	=	m	x	+	c
Output Label Prediction		slope weight	Input Feature		Intercept Bias

Error

Error / Residual → difference between Actual value & predicted Value

$$\text{Error}_i = y_i - (\hat{y}_i)$$

Performance across n sample predictions

Metrics

1. Mean Absolute Error (MAE) → Measures average magnitude without direction

$$MAE = \frac{1}{n} \sum_{i=1}^N |y_i - \hat{y}_i|$$

→ absolute difference

→ it is same exact units as target variable

2. Mean Squared Error(MSE)

Average of Squared Errors

$$MSE = \frac{1}{n} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

- error penalizes large ones more
- units are also squared

3. Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

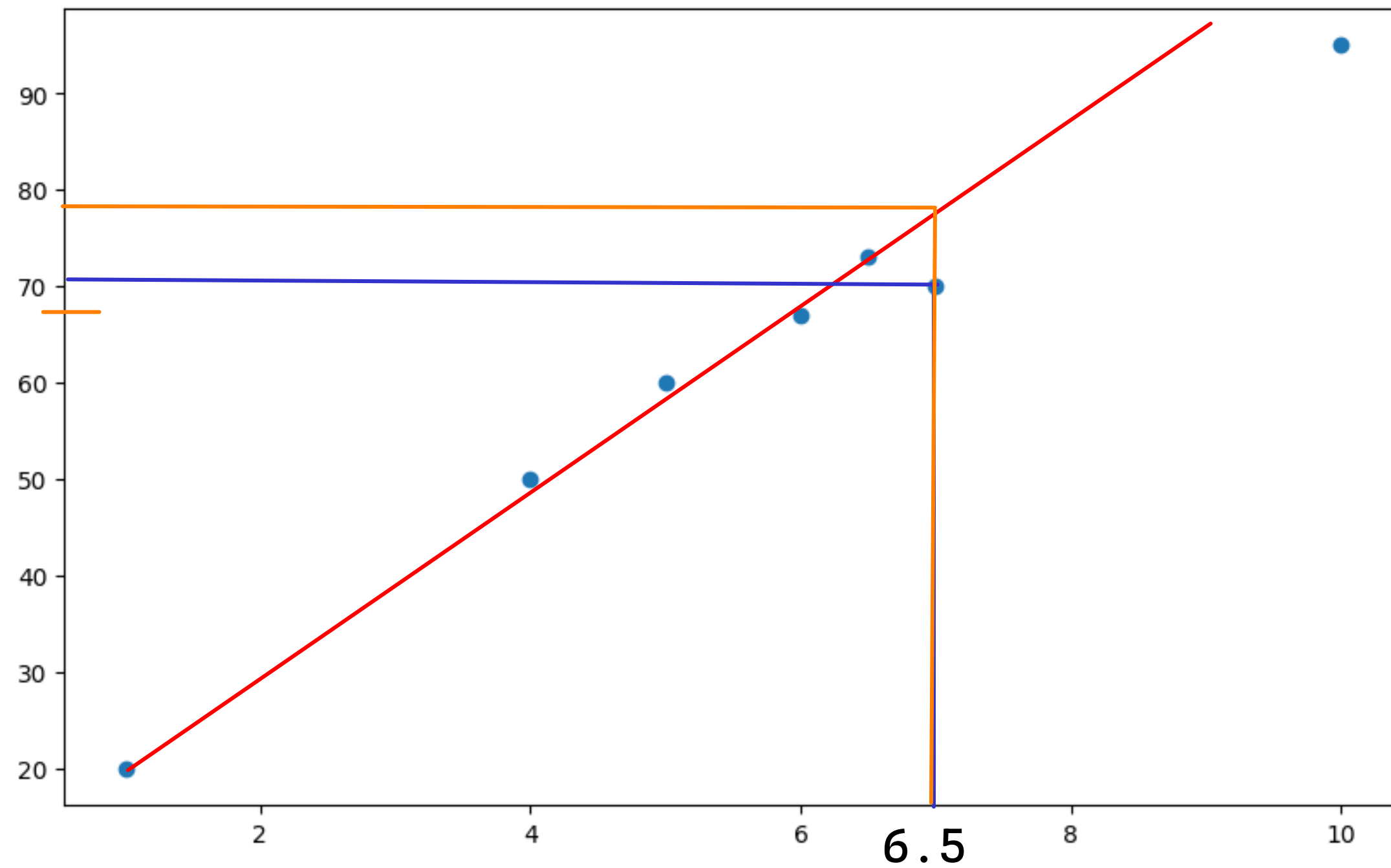
- It retains sensitivity to large errors, keeps original target units
- when it is desired to have exclude large errors; we use this

4. Coefficient of Determination R^2 Score

$$R^2 = 1 - \frac{SS_{res}}{SS_{total}} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

$\bar{y} \rightarrow$ mean of actual target values

73



69.2 → predicted
73 → actual value

loss/cost/ error

Lab: Perform Linear Regression for House Rent Prediction data

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Use the Features like Area, Bedrooms, Bathrooms, Floors

Label is Price